

**Thesis contract for topic:**  
**Physical touch of virtual objects**

---

UNIVERSITY OF SOUTHERN DENMARK

FACULTY OF ENGINEERING  
ADVANCED ROBOTICS TECHNOLOGY

---

*Author:*

Rokas Andrijauskas, roand21@student.sdu.dk

*Supervisor:*

Xiaofeng Xiong, xizi@mmmi.sdu.dk

# 1 Project Description

**Context** Recent advancements in computer hardware and software have enabled nearly photorealistic simulation of virtual worlds. Technologies such as augmented reality (AR) and virtual reality (VR) enhance the immersion of these simulated worlds by bringing the visuals into a third dimension. This is usually achieved through the use of a headset that overlays information over the user's field of view. Although the screens inside a VR headset are still flat, the portability and additional sensors allow for a natural way to look around the world, making the user feel like he is inside a new world. Similarly, headphones complement the visual experience by overlaying environmental sounds with artificial ones.

**Problem** A limitation of current VR technology is that it usually immerses only two senses: sight and hearing, but we, as humans, have five: sight, hearing, taste, smell, and touch. The thesis aims to address this limitation by incorporating the sense of touch into the computer-simulated virtual worlds.

**Solution proposal** The sense of touch is mediated by neurons that are embedded in the skin, the body's largest organ. Two potential approaches to overwriting this sense would be either by developing a brain computer interface that interacts with the brain region responsible for sensing or developing a full-body suit. Unfortunately, both of these methods require extensive research and resources that are beyond the scope of this thesis. Instead, the thesis will be focused on developing a portable exoskeleton that is designed for the hand, as it is the part of the body that interacts with the environment the most and has a high concentration of touch receptors.

**How is project delimited?** There are three distinct parts this project is comprised of: the design and manufacturing of mechanical exoskeleton, the implementation of controller and the creation of virtual world. The thesis will be done by a team of three students, each responsible for one of these parts. My role will be the development of a virtual world. The main objectives are:

- Research ways to simulate real-time soft-body object dynamics.
- Creation of a virtual environment with objects that have different material properties.
- Enabling interaction with the environment through the use of VR, 3D motion capture and the exoskeleton.

**Skills that will need to be learned** The engine with which the virtual world will be created and programming language it uses, will need to be mastered to a degree that allows for the completion of the objectives.

**What theories and technologies are in play?** 3D motion capture, game development, simulation.

**Overall timeline**